

Investigating Vection-Based Directional Guidance during Snowboarding Using a VR Simulator

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Abstract

Visual stimuli such as vection fields are known to unconsciously guide walking direction. However, their effects on other forms of locomotion remain largely unexplored. For example, snowboarding poses unique challenges for directional guidance because, unlike walking, the rider's stance is perpendicular to the direction of travel, and their gaze and board directions do not always align with it. This study examines the effects of three vection field presentation methods—head-relative, board-relative, and world-relative—on directional guidance during snowboarding using a VR simulator. Results showed an asymmetry between leftward and rightward visual stimuli, likely due to the body's front-back asymmetry. These findings suggest that visual stimuli may induce snowboard-specific patterns of directional guidance.

CCS Concepts

• **CSS Concepts**; • **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI;

Keywords

Snowboarding, Visual stimuli, Direction guidance, Vection, Self-motion Perception

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1 INTRODUCTION

Visual stimuli such as optical flow can unconsciously steer human walking direction by inducing vection, a self-motion perception

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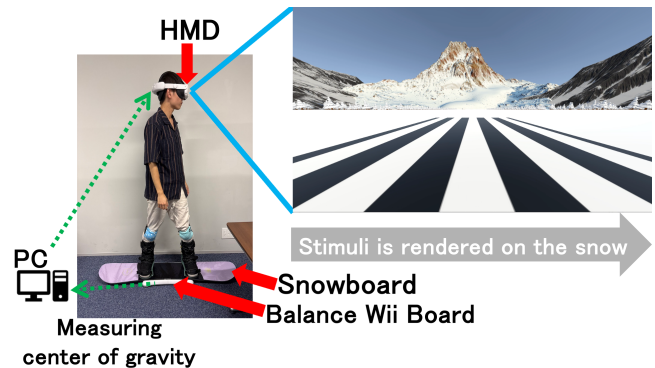


Figure 1: VR simulator components used in the experiment.

generated purely by vision [2, 4, 7, 9]. While this effect has been investigated for walking [3, 5, 8, 10], its potential to guide other forms of locomotion remains largely unexplored. This study focuses on snowboarding, a form of locomotion where shifts in the center-of-gravity change the travel direction and visual cues may also play a role. However, snowboarding poses unique challenges for directional guidance due to its posture and locomotion characteristics: the rider's stance is perpendicular to the direction of travel, and their gaze and board directions do not always align with it. In walking, these directions are typically consistent, particularly in controlled laboratory settings. These differences may lead to directional guidance characteristics that differ from those in walking.

This study investigated the effects of vection-based directional guidance during snowboarding. Visual stimulus patterns were set aligned with the rider's head, board, and global axes, as these directions differ from the travel direction during snowboarding. This paper presents results of experiments conducted using a VR snowboarding simulator.

2 SNOWBOARDING SIMULATING SYSTEM

To conduct the experiment under controlled and safe conditions, we developed a snowboarding simulator in a VR environment (Fig.

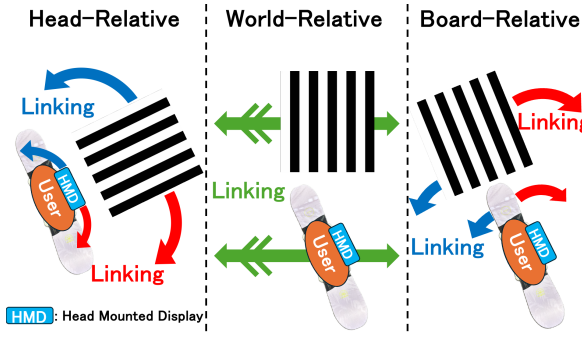


Figure 2: Three visual stimulus presentation conditions.

1). The system consists of three main components: (1) a VR environment, (2) a head-mounted display (HMD) for visual presentation, and (3) a Nintendo Balance Wii Board for detecting center-of-gravity shifts [1].

Participants wore snowboard boots mounted on a snowboard placed on a Balance Wii Board, with their ankles fixed to the board. In this simulation, the Y-axis coordinate on the Balance Board was used to reproduce left-right turning during snowboarding. When participants shifted their center-of-gravity forward or backward, the sensor detected the change in the Y-axis coordinate, and the user's travel direction in the VR environment changed to the right or left, respectively.

In the experimental course within the VR environment, trees were positioned along both edges, and a red goal line was displayed. Participants rode straight toward the goal line from the starting point with no obstacles to avoid.

3 EXPERIMENT

3.1 Visual Stimuli Using Vection Fields

The visual stimuli consisted of black-and-white stripes, appearing 2 meters wide and 3 meters deep to the participants. These stripes moved continuously in a fixed direction at a constant speed, either leftward or rightward. The vection field, which contained these visual stimuli, was presented on the ground.

We set three experimental conditions with different vection field presentation patterns, each positioned to align with different directions: head-relative, board-relative, and world-relative conditions (Fig. 2). In the head-relative condition, the vection field was displayed aligned with the participant's forward head orientation, with the flow direction perpendicular to it. In the board-relative condition, the field was aligned with the snowboard's forward orientation, with the flow direction perpendicular to the board's longitudinal axis. In the world-relative condition, the field was consistently displayed ahead toward the goal direction, with the flow direction perpendicular to it.

3.2 Experimental procedure

Sixteen university students (aged 20–24 years) without prior experience in snowboarding participated after providing informed consent. They wore snowboard boots on the board and vibration devices on their knees and ankles to provide vibrational feedback

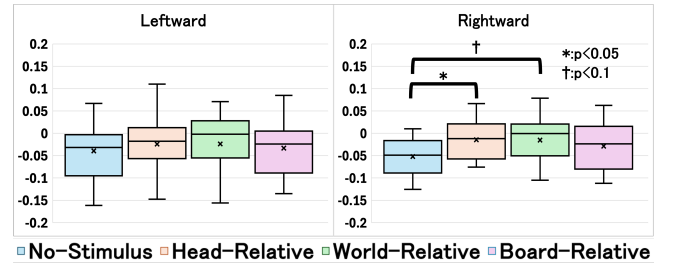


Figure 3: Average Y-axis center-of-gravity coordinates by leftward and rightward visual stimulus directions.

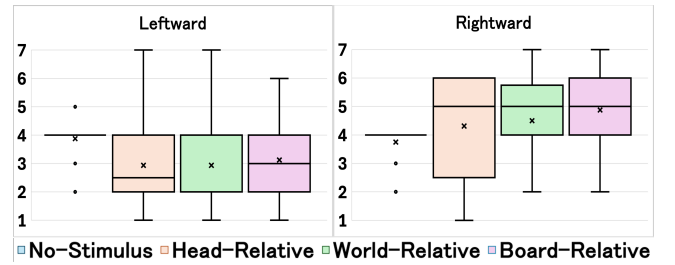


Figure 4: Subjective evaluation results of induction effect by Leftward and Rightward visual stimulus directions.

during gliding. After instructions on snowboarding in the simulator, they completed one practice trial without visual stimuli on a separate course from the main experiment. In the experiment, trials for rightward and leftward visual stimuli were conducted separately. For each direction, participants completed four trials: no stimulus, head-relative, board-relative, and world-relative conditions. Their order was determined using a Latin square design. Before each trial, participants were instructed to ride straight toward a red goal line. During each trial, participants' center-of-gravity coordinates were recorded. After each trial, they were asked, "Did you feel as if you were being pulled to either the left or right during this run?" Responses were given on a 7-point scale (1=strongly left 4=neutral, 7=strongly right).

4 RESULTS AND DISCUSSIONS

The directional guidance effect was evaluated using center-of-gravity data and questionnaire responses. The mean Y-axis value of the center-of-gravity for each participant and trial was calculated (positive=rightward, negative=leftward relative to the riding direction). Figure 3 shows that, overall, participants' center-of-gravity shifted more to the right for both stimuli compared to the no-stimulus condition. The results of repeated measure ANOVA showed no significant differences for the leftward stimulus, but did for the rightward stimulus ($p=0.019$). Bonferroni-adjusted pairwise comparisons showed that the head-relative ($p=0.008$) and world-relative ($p=0.078$) conditions produced significant and marginally significant rightward shifts, respectively, compared to the no-stimulus condition.

This asymmetry may be due to human balance control: when the ground appears to move, people lean in that direction to maintain

balance. Rightward stimuli naturally caused a rightward weight shift. However, for leftward stimuli, backward (i.e., leftward relative to their travel) leaning was corrected quickly because humans are generally more sensitive to backward leaning [6]. It keeps the center-of-gravity more forward (i.e., rightward) compared to the no-stimulus condition. Subjective results supported this: Figure 4 shows that participants felt pulled in the same direction as the visual stimulus, with marginal significant difference found in ANOVA. That is, they felt pulled to the left when leftward stimuli were presented.

The small difference between the head-relative and world-relative conditions may be because participants kept their heads facing the goal line as instructed. This likely aligned the stimulus with the goal, that is, with their gaze direction, in both conditions, making them feel similar. In the board-relative condition, the effect of visual stimuli was smaller than in the other two stimulus conditions. The board-relative stimuli followed the rider's actual direction, so it may help participants adjust to maintain a straight path.

5 FUTURE WORK

This study suggests that visual stimuli may induce snowboard-specific directional guidance, potentially influenced by posture and body asymmetry. While the precise impact of stimulus direction remains unclear, our findings provide new insights for designing snowboarding guidance interfaces. Future research should clarify these mechanisms and the effects of different stimulus directions.

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